

Q1) Fill the table with the correct answer (12 marks)

1	2	3	4	5	6	7	8	9	10	11	12

1) Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

- A) 0 and 127. B) 256 and 255. C) -128 and 128. D) 0 and 255

2) If 1110 0100 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

- A) -28. B) +228. C) -228. D) -74

3) Which of the following summation set the CF and SF?

- A) 7FF1h + 000Fh B) 7FFEh + 0001h C) FF00h + FF00h D) 8000h + 8000h

4) Which of the following statement is correct, if X DB 22H

- A) MOV [SP], 99H B) MOV BL, OFFSET X C) PUSH XD) none of them

5) What is the type of addressing for the following statement: MOV BX, [SI]

- A) Indexed B) indexed based C) direct D) based

6) If you want to compare the following two signed values AX = 123Fh, BX = B3F0h and put the value 1 in CL if AX < BX, you write:

- A) CMP AX, BX
JL L1
.....
L1: MOV CL, 1
B) CMP AX, BX
JA L1
.....
L1: MOV CL, 1
C) CMP AX, BX
JB L1
.....
L1: MOV CL, 1
D) CMP AX, BX
JG L1
.....
L1: MOV CL, 1

7) The value of CX after executing the following code is:

```
MOV AL, 0FFH
MOV CX, 5
L1: ADD AL, 1
    LOOPZ L1
```

- A) CX=0 B) CX=5 C) CX= 4 D) CX=3

8) What is the value of the AL register after executing the following code?

```
MOV AX, 0FF02H
MOV CL, 0FCH
XCHG AL, CL
ADD AL, 5
SUB AL, 1
```

- A) 01H B) 00H C) FFH D) None of them

9) Number of the times the instruction sequence below will loop before coming out of loop is

```
MOV AL, 00H
LI: INC AL
JNZ LI
```

- A) 0 B) 1 C) 256 D) infinite loop

Assume: SS=1000H, DS=7000H, SI=0020H, DI=0100H, X=0500H, BP=0300h

10) What is the physical address of value that will be stored in DH for the following instruction `MOV DX, X[SI+2]`

- A) 7522h B) 10522h C) 70522h D) 70523h

11) Calculate the physical address of the destination operand for the following instruction `MOV [BP + 40H], CL`

- A) 70340h B) 10340h C) 1340h D) 7340h

12) What is the value of the SP register after executing the following instruction, if the SP = FFFA4700 AX

```
DEC SP
PUSH BX
```

- A) FFF7H B) FFF8H C) FFFBH D) FFF9H

Q2) Trace the following codes and find the result of executing by filling the blanks. (25 marks)

```
1) MOV CX, 8
   MOV BL, 2
   MOV AL, 9
   L: MOV AH, 0
   DIV BL
   CMP AL, 0
   LOOPNZ L
```

AH=

AL=

BL=

CX=

```

2) MOV AX, 008AH
   MOV DX, F380H
   MOV BL, 2
   ADD AL, DL
   CMP DH, 2
   JL L
   DIV BL
   JMP L1
L: MUL BL
L1: HLT

```

AH=

AL=

DH=

DL=

```

3) X DW 733H, 1122H
   MOV AX, 4455H
   PUSH AX
   PUSH X+1
   POP AX
   POP BX

```

AH=

AL=

BH=

BL=

```

4) X DB 3, 5, 7
   INC X+1
   MOV AX, WORD PTR X+1
   NEG AX

```

AL=

AH=

CF=

SF=

PF=

AF=

```

5) X DW 5437H
   Y DW 28H
   MOV BX, OFFSET X
   ADD BX, 1
   MOV AL, [BX]
   MOV CL, [BX+1]
   ADD AL, CL
   INC BX
   MOV DL, [BX]

```

DL=

AH=

AL=

CL=

```

6) X DB 3, -5, 4
   MOV CX, 2
   MOV SI, 1
   L2: TESTX(SI), 81H
   JZ L1
   NOT(SI)
   L1: INCSI
   LOOP L2

```

X = _____

Q3)

a) Suppose we have a microprocessor with 24 bits address bus, what is the maximum memory capacity will be? (1 mark)

b) Write the equivalent assembly codes to the following C++ codes: (3 marks)

Note: x, y and s are unsigned byte variable.

```

if (x > y)
    s = s + x;
else
    s = s - y;

```

Q#1:[12]Circle the correct answer:

1. Data bus width of Intel 8086 microprocessor is

- (a) 16 (b) 24 (c) 20 (d) 32

2. The length of extended ASCII codes

- (a) 7 bits (b) 8 bits (c) 12 bits (d) 16 bits

3. One of the following is not part of execution unit of 8086 microprocessor

- (a) BP register (b) AX register (c) DS register (d) SP register

4. The equivalent code of NEG AX instruction is:

- a) XOR AX,AX
SUB AX,1
b) SUB AX,AX
ADD AX,1
c) OR AX,AX
ADD AX,1
d) XOR AX,FFFFH
ADD AX,1

5. A 16-bit address bus allows access to a memory of capacity

- (a) 128KB (b) 64KB (c) 256KB (d) 512KB

6. If BX=8321h, CF=1 and CL=5 after execution ROR BX,CL the:

- a) BX=0C19H and CF=1
b) BX=0C19H and CF=0
c) BX=1832H and CF=0
d) BX=1832H and CF=1

7. One of the following control transfer instruction is not used with signed number

- a) JGE (b) JBE (c) JL (d) JLE

8. After execution of the following code register AL contains

```
MOV AL, 'a'
ADD AL, 2
AND AL, 1101 1111b
```

- a) AL='c' (b) AL='C' (c) AL='a' (d) AL='A'

10. If the SP register contains the decimal value 340, what will be its value after executing the following program segment?

```
PUSH AX      MOV BX, 25
PUSH BX      MUL BX
PUSH DX      POP BX
```

- a) 344 b) 338 c) 342 d) 336

11. Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

- a) 0 and 127, b) 256 and 255
c) -128 and 128, d) 0 and 255

12. If 1110 0100 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

- a) -28, b) +228,
c) -228, d) -74,

Q2: [15 points] Trace the following programs and determine the final value for each register:

a) MOV AL, 05H
NEG AL
MOV BL, 02H
AND BL, AL

BL=	AL=	PF
2	FB	0

b) MOV BL, 07H
SUB BL, 0EH
MOV DL, 02H
MOV AL, FFH
ADD AL, 1
JCL
MOV CL, FFH
JNC LL
L: MOV CL, 00
LL: ADC DL, 1

AL=	BL=	CL=	DL=
00	F9	00	02

c) MOV CL, 03H
MOV AX, 127
ROR AX, 01H
JCL
ADC CL, 02H
L: HLT

AX=	CL=	CF=	ZF=	PF=
803F	03	1	0	1

a) MOV AX,35H
 MOV BX,39H
 PUSH AX
 PUSH BX
 ADD AX,BX
 DAA
 MOV DX,AX
 POP AX
 POP BX

AX=	BX=	DX=
35	39	39

b) MOV AL,14
 MOV AH,0
 MOV BL,17
 DIV BL

AL= 082 AH= 00

Question #2

Suppose we have a microprocessor with 24 bits address bus:

a) What is the maximum memory capacity will be?

max memory = 2^{24}

b) For the following instruction MOV [BX+76Dh],AL

1- What is the addressing mode used?

Base addressing mode with displacement

2- Calculate the physical address of memory location assuming that DS=8CFBh and BX=4000h

ph address = $reg \times 10h + off$

[5 points]

phy add = $8CFBh + 10h + 4000h$
 $= 90FB0$

10

بسم الله الرحمن الرحيم

X	DB	5,6,-20,30,40,1,-4,-6
---	----	-----------------------

Y	DW	8 DUP(0)
---	----	----------

Loop

J

HLT

[illegible]

4/4

[4 points]

Question #4

Choose the correct answer:

8

1. Address bus width of Intel 8086 microprocessor is

☒ (a) 16

(b) 24

(c) 20

(d) 32

2. The length of ASCII codes

☒ (a) 7 bits

(b) 8 bits

(c) 12 bits

(d) 16bits

3. One of the following is a part of bus interface unit of 8086 microprocessor

(a) BP register (b) AX register

☒ (c) DS register

(d) SP register

4. The equivalent code of NOT AX instruction is:

a) XOR AX, FF
ADD AX, 1

b) SUB AX, AX
ADD AX, 1

c) XOR AX, AX
ADD AX, 1

☒ (d) XOR AX, FFFFFH

5. A 34-bit address bus allows access to a memory of capacity

(a) 16MB

(b) 64MB

(c) 4GB

☒ (d) 16GB

6. One of the following control transfer instruction is not used with unsigned number

☒ (a) JGE

(b) JBE

(c) JA

(d) JAE

7. The stack pointer always points to a(n) ... location.

(a) full

(b) empty

☒ (c) byte

(d) None of the above

8. If the SP register contains the decimal value 340, what will be its value after executing the following program segment?

PUSH AX

MOV BX, 25

PUSH BX

MUL BX

PUSH DX

POP BX

(a) 344

(b) 338

(c) 342

☒ (d) 336

9. Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

(a) 0 and 127

(b) 0 and 128

(c) -128 and -127

☒ (d) 0 and 255

10. If 11111110 is the 8-bit 2's complement representation of a number N, what is the decimal value of N

☒ (a) -2

(b) 254

(c) -3

(d) 1

QUESTION #3

a) Write an equivalent Assembly code for the following statement

If $X > 0$ res = $((X*10) - Y) - Z$ else res = $X - 2$

Note: X, Y and Z are byte sizes, value unsigned; res is word size.

X DB 2
Y DB 4
Z DB 6
res DW ?
mov cx, 10
mov AL, X
cmp AL, 0
JG L1
mul cx

ADD AX, Y
SUB AX, Z
Jmp L2
L1: mul BX, AX
mov res, AX
L2: HLT

b) Write an equivalent of the following instructions:

LOOP L
L1: mov AL, H
L1: sub AL, 1
dec cx
jnz L1

NOT BL
Neg BL
Dec BL
SUB AL, BL
Neg BL
ADD AL, BL

c) Write instructions to test bit-2 in register AL, if it is one reset it otherwise invert all bits in register AL.

AND AL, 04
JNE L1
NOT AL
Jmp L2
L1: AND AX, FD
L2: HLT

d) Write an assembler codes to push into the stack even numbers from 0-100

mov cx, 100
mov SI, 0
L1: AND AX, 01
JNZ L1
push X[SI]
INC SI
DEC CX
JNZ L1
Jmp L2
L1: Jmp L2
L2: HLT

Al-Balqa Applied University Faculty of Engineering Technology Computer Engineering Department 33	 Microprocessors and Assembly Language MID Exam	Date:
		Instructor Name:
		Student Name:

QUESTION #1

Find the output of the following program segments:

a) <pre> X DB 1H,4H,3H Z DW 1030H MOV AL,[X+2], AL = FD MOV AH,30H, AH = 30 AND Z,AX, Z = 1030H NEG Z HLT </pre> AX = 30FD ; Z = EF0F	b) <pre> X DD 8ACF0123H,0FB432DH MOV SI,0 MOV BX,OFFSET X MOV AL,[BX], AL = 23 INC BX INC BX MOV AH,[BX], AH = CF MOV DX,[BX+3], DX = FB43 XCHG DL,X[SI], DL = 23 < 43 </pre> AX = CF23 ; DX = FB43 ; X = 43H
c) <pre> MOV CL,04H ; CL = 04 MOV AL,7, AL = 07 ROR AL,CL, AL = 9F JNC L ADC CL,01H, CL = 05 JMP L2 L: MOV DL,AL AND AL,0FH L2: HLT </pre> AL = 9FH ; DL = 05H ; ZF = 0	d) <pre> X DW 501FH MOV AX,35H ; AX = 35 MOV CL,8 ; CL = 08H MOV BX,39H ; BX = 39H PUSH AX PUSH X INC X, 5020 POP BX, BX = 501F SHR AX,CL, AX = 0000 POP DX, DX = 0035 AND DX,BX, DX = 0015 </pre> AX = 0000H ; BX = 501F ; DX = 0015 ; X = 5020H

QUESTION #2

Assume that you have a microprocessor called ASSEM_MICRO, the address bus size equal 14 bits the data bus size 8 bits and the register size is 12 bits; CS=600H; IP=50H; Calculate the physical address for the instruction pointed by IP

$$\begin{aligned}
 PA &= 16 \times 2^6 + \text{offset} \\
 &= 600 \times 16 + 50 = 24050
 \end{aligned}$$



Faculty of Engineering Technology

Microprocessors and Assembly Language (Mid Exam)

STUDENT NAME: أحمد محمد العتيبي

Marks(40)

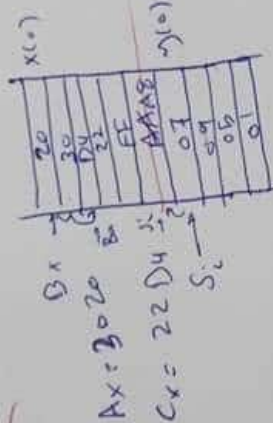
Q1: Multiple Choice Questions: (18 pts)

- 1- The 16 bit flag reg of 8086 microprocessor is responsible to indicate:
☒ A. the condition of result of ALU operation ☐ B. the condition of memory ☐ C. the result of subtraction
- 2- The CF is known as:
☐ A. condition flag ☒ B. carry flag ☐ C. common flag
3. The SF is called as _____
☐ A. service flag ☒ B. sign flag ☐ C. single flag
4. The IP is _____ bits in length
☐ A. 8 bits ☐ B. 4 bits ☒ C. 16 bits
5. To exchange 2 bytes stored at CL and DL regs:
☐ A. xchange DL,CL ☒ B. xchg DL,CL ☐ C. MOV DL,CL
6. The result of MOV AL, 65 is to store
☐ A. store 0100 0010 in AL ☐ B. store 42H in AL ☒ C. store 0100 0001 in AL
7. Which of the following instruction is not valid?
☐ A. MOV AX, BX ☒ B. MOV DS, 5000H ☐ C. MOV SI, 5000H
8. In PUSH instruction, after each execution of the instruction, the stack pointer
☐ A. incremented by 1 ☐ B. decremented by 1 ☒ C. incremented by 2
9. Which of the following instruction is correct?
☐ A. MOV X, AX ☒ B. LEA BL, X ☒ C. MOV DX, [AX]
10. Imul instruction treats the FCH value as :
☐ A. 4D ☒ B. 4D ☐ C. 1111110B

Q3) Write the type of addressing mode of the following code example (5PTS.)

- MOV BL, 24H → Immediate addressing mode
- MOV AL, [3000H] → Direct addressing mode
- MOV BX, [SI] → Index addressing mode
- MOV CX, [BX+DI] → Base Index addressing mode
- MOV CX, ARRAY[DI] → Indirect addressing mode

AX = 2
A8



Q4) Trace the following: (10 PTS.)

- 1) X DW 3020H, 22D4H, 0AAFFH
Y DB 7, 9, 5, 1
MOV BX, OFFSET X
MOV AX, [BX]
ADD BX, 2
MOV CX, [BX]
MOV SI, OFFSET Y
SUB SI, 1
SUB [SI], 2

AX = 3020H

CX = 22D4H

DX =

X = 3020H

2) MOV BL, -122

ADD BL, 7

BL = 8D H

SF = 1

PF = 1

OF = 0

$$\begin{array}{r} -122 \\ + 7 \\ \hline -115 \end{array}$$

3) MOV AL, 18

MOV BL, 10

MUL BL

DEC BL

AX = B4 H

BL = 9 H

$$\begin{array}{r} AX = 18 \times 10 \\ = 180 \end{array}$$

$$\begin{array}{r} 10000110 \\ + 10001101 \\ \hline 10001011 \\ 8D \end{array}$$

$$\begin{array}{r} 264 \\ + 32 \\ \hline 296 \end{array}$$

$$\begin{array}{r} 128 \\ + 16 \\ \hline 144 \end{array}$$

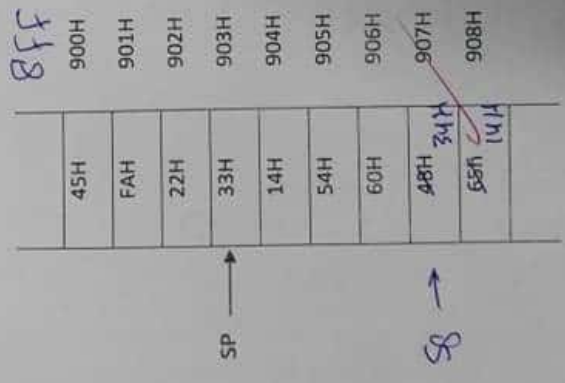
$$\begin{array}{r} 8421 \\ 0000 \\ 0000 \\ 1010 \end{array}$$

$\frac{2}{98}$
 11. What is the possible decimal value could the 8-bit number 9Eh represent?
 A) 158 B) -158 C) -98

12. The effect of the following instructions on the ax register is:
 push ax
 add ax, 4
 pop bx
 mov cx, ax
 push bx
 pop ax
 A) leave it with its original value B. add 4 to it C. clear it

a) x = 5
 a/x = 9
 b = 9

Q2) Write suitable instructions to do the following (7 pts.)
 A. If sp=903h, INCREMENT the word pointed by sp in the stack, then store it back in stack location address 907h



$Ax = (433H) + 1434H$

B. After executing branch a, what is the stack after applying this code :
 Push 1430H
 Push 6589H
 SP = 903H

b) Complete the following table:

Instruction	Result	CF	ZF	SF	OF	PF	AF
MOV BH, 125	BH = 125	0	0	0	0	0	0
ADD BH, 6	BH = 131	0	0	0	0	0	0
SUB BH, BH	BH = 000	0	1	0	0	1	1
NEG BH	BH = 000	0	1	0	0	1	1

Third Question (12 marks):

a) (5 marks) For the following stack memory, fill-in the blanks with the correct answer:

1) SUB SP, 4
PUSH 205Bh
SP = 014Ch
Show the results in the stack blocks

SP	0500:014Ch
205B	0500:014Dh
CD	0500:014Eh
FE	0500:014Fh
FF	0500:0150h
49	0500:0151h
DD	0500:0152h
39	0500:0153h

2) POP DX
POP AX
DX = 205Bh
AX = FECDh

3) AND AX, 5
PUSH AX
AX = 0003h
Show the results in the stack blocks

SP = 014Eh

b) (7 marks) Consider the following initial register values: DS=000h, ES=3000h, SS=600h, BP=1009h, and the initial memory locations values are as shown in the figure. Answer the questions below:

1) For the instruction MOV SI, [BP-1], calculate the physical address needed to read the data from memory - if any:
Logical Address = 0600:100A
Physical Address Calculation:

$$\text{physical add} = 0600 \times 10h + 100A = 700Ah$$

SS →	7009h	24
	700Ah	DE
	700Bh	CC
	700Ch	09
DS →	C00Eh	12
	C00Fh	4E
	C0ECh	11
	C0Eh	80
ES →	C0E2h	07

Al-Balqa Applied University		Date: <u>29.5</u>
Faculty of Engineering Technology		Instructor Name: <u>31.5</u>
Computer Engineering Department		Student Name: <u>[Redacted]</u>

First Question (10 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
<u>B</u>	<u>A</u>	<u>A</u>	<u>D</u>	<u>C</u>	<u>X</u>	<u>A</u>	<u>D</u>	<u>A</u>	<u>B</u>

Consider the following code: Answer questions 1-3

```
X DB 1, 2, 3, 4, 5, 6, 7, 8
Z DW 0ABCDh
```

```
MOV AL, X-1
AND byte PTR Z, AL
NEG Z
```

```
inst#1
inst#2
inst#3
```



8

Q1) The value of AL after executing (inst#1) is:

- a) 1 b) FFh c) 8 d) Error

Q2) The value of memory location Z after executing (inst#2) is:

- a) 0600h b) 0ABCDh c) 0ABCDh d) 0CCABh

Q3) The value of Z after executing (inst#3) is:

- a) 5434h b) 5433h c) 3454h d) 3354h

Q4) What is the value of registers AX and BX after executing the following code are:

```
MOV AL, 10h
MOV BX, 1244h
XCHG AX, BX
```

- a) Ax = 10h b) Ax = 01A4h c) Ax = 12A4h d) Ax = 12A4h
 BX = 12A4h BX = 120B BX = 010B BX = 0002h

Consider the following data segment memory locations, answer questions from 5-8

0700:0102h	<u>20</u>	← X
0700:0103h	FF	
0700:0104h	4B	
0700:0105h	23	
0700:0106h	<u>1F</u>	← Y
0700:0107h	78	
0700:0108h	67	
0700:0109h	CD	

Q5) The value of BX after executing the following instruction is:

- a) 1Fh b) 0700:0106 c) 0106h d) 0700h

Q7) The value of DH after executing the following instruction is:

- a) 4Bh b) 0104 c) 0700:0104 d) Error

2) The value of DI after executing the instruction in 1 is

DI: ~~2A 2A 2A~~ D8H

3) The value of DL after executing the following instruction is:

MOV DL, ~~05~~ [DI+1]
DL: AEH

4) Give an instruction on each of the following addressing modes:

1- Direct addressing mode: MOV AL, 3

2- Based with displacement addressing mode: MOV AH, [BP-1]

3- Based indexed addressing mode: MOV BX, [BP+SI]

Fourth Question (4 marks): Write an equivalent Assembly code for the following equation:

$$res = ((X-10) * Y) / Z$$

Note: X, Y and Z are byte sizes

X DB 2
Y DB 5
Z DB 2
res DB ?

4

3.5

~~MOV AL, X~~
MOV AL, X
MOV BL, Y
MOV CL, Z
add AL, 10]
MUL BL]
div CL]
MOV res, AX
HLT

Q8) The value of X after executing the following instruction is

LEAD I, X

MOV [DI], 15

c) 15h

a) 10h

b) 15

DI = 0102h

d) 0Fh

Q9) Consider the following instructions:

arr DB 3 dup(?)

MOV arr(0), 12h

MOV arr(1), 2

MOV arr(2), 19h

The values of arr after executing the code are:

a) 12, FE, F9

b) 12, -2, F9

c) 12, 0Fh, 0Fh

d) Error

Q10) The instruction that is equivalent to the following operation is:

MOV AX, 0

a) AND AX, AX

b) XOR AX, AX

c) OR AX, AX

d) NOT AX

Second Question (14 marks):

a) Trace the following codes and fill-in the spaces:

1) MOV AL, 253
CMP AL, 0
JL L1
MOV CL, 0
JMP L2
L1: MOV CL, 1
L2:

CL = 1 H

2) MOV AL, -9
MOV BL, 2
IDIV BL

AX = FFFC H

3) MOV AL, 12B
MOV BL, 8
MUL BL

AX = 0400h H

4) Consider that CF = 1

MOV AX, 1
MOV BX, 1F3Dh
SBB AX, BX

AX = F0C8 H

5) Consider that: 'a' = 61h, 'A' = 41h
ARR DB 'abAab'

MOV SI, 0
MOV CL, 5
L2: CMP ARR[SI], 61h
JNE L1
SUB ARR[SI], 20h
L1: INC SI
DEC CL
CMP CL, 0
JAL L2
JZ Above

ARR = 'A6AAID'

6) Consider that the logical address for X is 0700:0114h

X DB 12, -3, 8, -5
Y DB 0

LEA BX, X
L2: TEST [BX], 80h
JZ L1
INC Y
L1: INC BX
CMP BX, OFFSET X+4
JNZ L2

BX = 0118 H
Y = 1 H